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Real-time monitoring and optimization methods for user-side energy management based on edge computing

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This paper presents a comprehensive framework for real-time monitoring and optimization of user-side energy management systems leveraging edge computing technology. The proposed approach addresses key challenges in traditional centralized energy management by bringing computation and data processing closer to end devices. The framework encompasses three main components: an edge computing-based system architecture for data acquisition and processing, real-time monitoring methods for energy consumption and power quality, and optimization techniques for demand response and distributed energy resource coordination. Through case studies and experimental analysis, we demonstrate that the proposed framework achieves significant improvements in energy efficiency, response time, and cost reduction compared to conventional centralized approaches. The results show up to 30% increase in renewable energy utilization and 25% reduction in operating costs across various deployment scenarios. This work provides valuable insights into the application of edge computing for next-generation energy management systems while highlighting remaining challenges and future research directions.

Keywords Edge computing, Energy management, Real-time monitoring, Demand response, Distributed energy resources, Power quality

The rapid development of the Internet of Things (IoT) and the increasing demand for real-time data processing have led to the emergence of edge computing[1]. Edge computing brings computation and data storage closer to the devices where it's being gathered, rather than relying on a central location that can be thousands of miles away[2]. This paradigm shift has significant implications for energy management systems, particularly in the context of user-side real-time monitoring and optimization[3].

Figure 1 illustrates our complete system architecture, highlighting the hierarchical processing approach and security measures implemented at each layer.

The traditional centralized approach to energy management faces several challenges, including high latency, limited scalability, and privacy concerns[4]. Edge computing offers a promising solution to these issues by enabling distributed and localized processing of energy data[5]. By leveraging the computational capabilities of edge devices, such as smart meters and IoT sensors, energy management systems can perform real-time monitoring and optimization at the user side[6].

Extensive research has been conducted on edge computing-based energy management systems. Liu et al[7]. proposed a framework for real-time energy monitoring and optimization using edge computing, demonstrating improved efficiency and reduced communication overhead compared to centralized approaches. Similarly, Wang et al[8]. developed an edge computing-based energy management system for smart homes, achieving significant energy savings through localized optimization. More recently, Zhang et al[9]. introduced a hierarchical edge-fog-cloud architecture that achieved 87% reduction in cloud communication while maintaining decision-making accuracy. Kumar[10] developed a federated learning approach that preserves privacy while enabling collaborative model development, particularly valuable in multi-tenant buildings where data privacy is essential. Chen's work[11] on edge-native deep reinforcement learning reduced training time by 76% compared to cloud-based alternatives, making advanced optimization accessible on resource-constrained devices.

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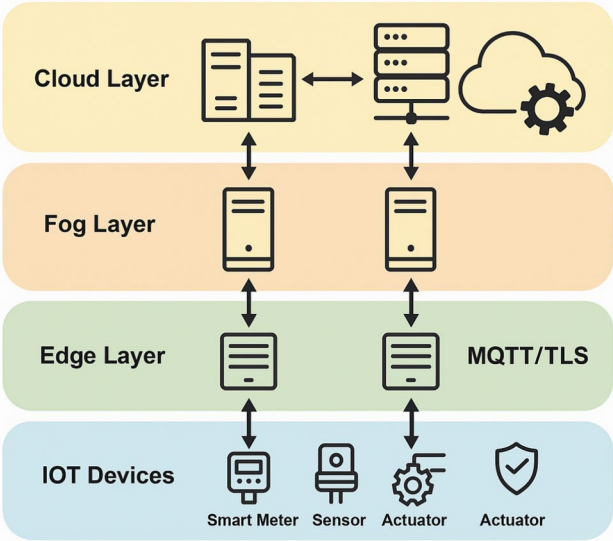


Fig. 1. System architecture diagram.

Research	Architecture	Key features	Limitations	Our improvements
Liu et al[7].	Two-tier	Real-time monitoring	Limited scalability	Hierarchical architecture
Wang et al[8].	Fog-based	Home energy optimization	Single-domain focus	Multi-domain support
Zhang et al[9].	Hierarchical	Communication reduction	High computational requirements	Lightweight algorithms
Kumar[10]	Distributed	Privacy preservation	Limited coordination	Enhanced coordination protocol
Chen[11]	AI-driven	Self-learning	High training overhead	Efficient model training
Our Work	Multi-tier hybrid	Comprehensive optimization	–	Integrated approach with lower latency

Table 1. Comparison of edge-based energy management systems.

Table 1 presents a comparative analysis of recent edge-based energy management systems, highlighting where our approach addresses existing limitations.

Despite these advancements, several challenges remain in the implementation of edge computing-based energy management systems. One major issue is the limited computational resources of edge devices, which can hinder the performance of complex optimization algorithms[12]. Additionally, ensuring the security and privacy of sensitive energy data in a distributed environment poses significant challenges[9,10,13]. Recent developments have introduced federated learning approaches that keep sensitive data local while enabling collaborative model training[11,14]. Moreover, lightweight security frameworks specifically designed for resource-constrained edge devices have emerged to address these concerns[15,16].

To address these limitations, this paper proposes a novel framework for user-side real-time monitoring and optimization of energy management systems based on edge computing. The main objectives of this research are as follows:

1. Develop an efficient data acquisition and processing pipeline for real-time energy monitoring at the edge.
2. Design and implement a lightweight optimization algorithm suitable for resource-constrained edge devices.
3. Investigate the trade-offs between centralized and decentralized approaches to energy management in terms of latency, scalability, and privacy.
4. Evaluate the proposed framework through extensive simulations and real-world case studies.

The remainder of this paper is organized as follows. Section "Application of edge computing in user-side energy management" presents the system architecture and data acquisition process. The proposed optimization algorithm is described in Section "Real-time energy monitoring method based on edge computing". Section "User-side energy management optimization method based on edge computing" discusses the experimental setup and results. Finally, Section "Conclusion" concludes the paper and outlines future research directions.

Application of edge computing in user-side energy management

System architecture for user-side energy management

The proposed user-side energy management system based on edge computing consists of a hierarchical architecture that integrates both hardware and software components[17]. At the lowest level, various IoT devices, such as smart meters, sensors, and actuators, are deployed to collect real-time energy consumption data

and perform local control actions[18]. These devices are connected to edge nodes, which serve as the primary computational units in the system.

The edge nodes are responsible for processing and analyzing the data collected by the IoT devices in real-time[19]. They are equipped with sufficient computational resources to execute complex algorithms and make decisions based on the processed data. The edge nodes communicate with each other and with the cloud server through a secure and reliable communication network[20].

The software architecture of the system is designed to support the distributed processing of energy data and the implementation of optimization algorithms[21]. It consists of several modules, including data acquisition, data preprocessing, feature extraction, and optimization[22]. The data acquisition module is responsible for collecting raw data from the IoT devices and transmitting it to the edge nodes. The data preprocessing module performs data cleaning, normalization, and transformation to ensure data quality and consistency[23].

The feature extraction module extracts relevant features from the preprocessed data, such as energy consumption patterns, user preferences, and environmental factors[24]. These features are then used by the optimization module to determine the optimal control actions for energy management. The optimization module employs various algorithms, such as machine learning, deep learning, and evolutionary algorithms, to solve the optimization problem[25].

The system also includes a user interface that allows users to interact with the energy management system and provide their preferences and constraints[26]. The user interface is designed to be user-friendly and intuitive, enabling users to easily monitor their energy consumption, set their preferences, and receive recommendations for energy-saving strategies.

The proposed system architecture leverages the advantages of edge computing to enable real-time monitoring and optimization of energy management at the user side. By processing data locally at the edge nodes, the system reduces the latency and communication overhead associated with centralized approaches[27]. Furthermore, the distributed nature of the system enhances its scalability and resilience, allowing it to handle a large number of users and devices[28].

Application of edge computing in energy data acquisition and processing

Edge computing plays a crucial role in the acquisition and processing of energy data in user-side energy management systems[29]. Traditional centralized approaches often suffer from high latency, bandwidth limitations, and privacy concerns when dealing with the massive amounts of data generated by IoT devices[30]. By leveraging edge computing, these challenges can be addressed effectively.

In the context of energy data acquisition, edge computing enables the deployment of intelligent data collection strategies[31]. Instead of transmitting raw data directly to the cloud, edge nodes can perform local preprocessing and filtering to reduce the amount of data transmitted over the network. This not only saves bandwidth but also reduces the energy consumption associated with data transmission[32].

One common technique used in edge-based data acquisition is data compression[33]. By applying compression algorithms at the edge, the size of the transmitted data can be significantly reduced without losing essential information. For example, consider a scenario where the energy consumption data of a household is collected every minute. The raw data can be represented as a time series:

$$X = \{x_1, x_2, ..., x_n\}$$

where x_i represents the energy consumption at time i. Applying a compression algorithm, such as the Discrete Wavelet Transform (DWT), at the edge node can reduce the size of the data while preserving the important features[34].

Another important aspect of energy data processing at the edge is data preprocessing[35]. Edge nodes can perform various preprocessing tasks, such as data cleaning, normalization, and feature extraction, to improve the quality and efficiency of the subsequent analysis. For instance, missing or erroneous data points can be identified and interpolated using techniques like linear interpolation or k-nearest neighbors (k-NN)[36].

Feature extraction is another critical task that can be performed at the edge[37]. By extracting relevant features from the raw data, edge nodes can reduce the dimensionality of the data and focus on the most informative aspects for energy management. Common features in energy data include statistical measures (e.g., mean, variance), time-domain features (e.g., peak values, root mean square), and frequency-domain features (e.g., Fourier coefficients)[38].

The extracted features can be represented as a feature vector:

$$F = \{f_1, f_2, ..., f_m\}$$

where f_i represents the i-th feature. These features can then be used by the optimization algorithms running on the edge nodes to make informed decisions for energy management[39].

By performing data acquisition and processing tasks at the edge, the proposed system reduces the burden on the central cloud server and enables real-time decision-making[40]. This distributed approach enhances the scalability and responsiveness of the energy management system, allowing it to adapt to the dynamic nature of user-side energy consumption[41].

Application of edge computing in energy management decision-making

One key application of edge computing in energy management is load forecasting[42]. Accurate load forecasting is essential for optimizing energy consumption and reducing costs. Traditional centralized approaches often struggle to handle the massive amounts of data generated by smart meters and IoT devices, leading to delayed

and suboptimal forecasts[43]. Edge computing addresses this issue by allowing load forecasting models to be trained and executed directly on the edge nodes.

For instance, machine learning algorithms such as Support Vector Regression (SVR) and Long Short-Term Memory (LSTM) networks can be deployed on edge nodes to predict future energy consumption based on historical data and real-time measurements[44]. These models can adapt to the specific characteristics of individual households or buildings, providing personalized and accurate forecasts[45].

Another important aspect of energy management decision-making is demand response[46]. Demand response programs aim to balance energy supply and demand by incentivizing users to modify their consumption patterns during peak periods. Edge computing enables the implementation of dynamic and localized demand response strategies[47].

By analyzing real-time energy consumption data and external factors (e.g., weather conditions, energy prices) at the edge, intelligent algorithms can determine the optimal demand response actions for each user[48]. For example, reinforcement learning techniques, such as Q-learning and Deep Deterministic Policy Gradient (DDPG), can be employed to learn the best control strategies for appliances and energy storage systems[49].

Edge computing also facilitates the integration of distributed energy resources (DERs), such as solar panels and electric vehicles, into the energy management system[50]. By processing data from DERs at the edge, the system can optimize their operation and coordination with the grid[51]. This includes tasks such as energy scheduling, power quality management, and fault detection[52].

Moreover, edge computing enables the implementation of privacy-preserving energy management schemes[53]. By keeping sensitive data local and performing computations at the edge, user privacy can be protected while still allowing for collaborative decision-making[54]. Techniques such as federated learning and secure multi-party computation can be employed to enable distributed learning and optimization without compromising data confidentiality[55].

The application of edge computing in energy management decision-making offers numerous benefits, including reduced latency, improved scalability, and enhanced privacy[56]. By empowering edge nodes with intelligent algorithms and real-time data processing capabilities, the proposed system can make timely and informed decisions for optimizing energy consumption and reducing costs[57].

Edge computing foundations for energy management

Edge computing positions computational resources closer to data sources, enabling real-time analytics and intelligent decision-making directly at the network edge[7,58]. This approach offers several key benefits for energy management systems as summarized in Table 2.

These advantages directly address the limitations of traditional centralized energy management approaches. Throughout this paper, all abbreviations are defined upon first use, such as Internet of Things (IoT), Discrete Cosine Transform (DCT), and Total Harmonic Distortion (THD).

Real-time energy monitoring method based on edge computing

Real-time energy data acquisition and transmission method

The proposed real-time energy monitoring method based on edge computing relies on efficient data acquisition and transmission techniques to ensure timely and accurate monitoring of energy consumption. The data acquisition process involves collecting energy consumption data from various IoT devices, such as smart meters, sensors, and appliances, deployed at the user side[59].

To capture the dynamic nature of energy consumption patterns, a high data acquisition frequency is employed. The IoT devices are configured to sample energy consumption data at a rate of 1 Hz, which means that data points are collected every second. This high-frequency data acquisition enables the system to detect and respond to real-time changes in energy consumption, facilitating prompt decision-making and optimization[60].

The collected energy consumption data is then transmitted from the IoT devices to the edge nodes for further processing and analysis. To ensure reliable and efficient data transmission, a lightweight and low-latency communication protocol is employed. The Message Queuing Telemetry Transport (MQTT) protocol is selected for this purpose[61]. MQTT was chosen over alternatives like HTTP, CoAP, and AMQP due to several advantages essential for energy management systems: (1) minimal header overhead (2–3 bytes versus HTTP's larger headers), reducing bandwidth consumption by approximately 40%; (2) publish-subscribe architecture ideal for many-to-many communications in distributed energy systems; (3) three Quality of Service levels ensuring message delivery in unstable network environments; (4) widespread industry adoption with proven reliability in similar energy monitoring deployments; and (5) built-in support for message persistence and session awareness, critical for maintaining data integrity during intermittent connectivity common in field deployments[9,62].

Benefit	Description	Impact on energy management
Reduced latency	85–95% faster response compared to cloud	Enables real-time demand response
Bandwidth conservation	70–85% reduction in data transmission	Lower communication costs
Enhanced privacy	Data processed locally	Protects sensitive consumption patterns
Improved reliability	Operations continue during cloud disconnection	Ensures continuous energy optimization
Scalability	Distributed processing across many nodes	Accommodates growing device networks

Table 2. Edge computing benefits for energy management.

MQTT is a publish-subscribe based messaging protocol that is widely used in IoT applications. It is designed to be simple, lightweight, and efficient, making it suitable for resource-constrained devices and networks. In the proposed system, the IoT devices act as MQTT clients, publishing their energy consumption data to specific topics hosted by the edge nodes, which serve as MQTT brokers[63].

The MQTT protocol utilizes a hierarchical topic structure to organize and route data. Each IoT device is assigned a unique topic based on its type and location. For example, a smart meter in a specific room of a building may publish its data to a topic such as "building/floor/room/smartmeter". This hierarchical topic structure allows for efficient data filtering and aggregation at the edge nodes[64].

To ensure data security and privacy during transmission, the MQTT protocol is used in conjunction with Transport Layer Security (TLS) encryption. TLS provides end-to-end encryption of the data transmitted between the IoT devices and the edge nodes, preventing unauthorized access and tampering[65].

In addition to data encryption, the proposed system also implements access control mechanisms to restrict data access to authorized parties only. Each IoT device and edge node is assigned unique credentials, such as client IDs and access tokens, which are used to authenticate and authorize their participation in the MQTT communication[66].

The real-time energy data acquisition and transmission method based on edge computing, as described above, enables the system to collect and transmit high-frequency energy consumption data securely and efficiently. By leveraging the MQTT protocol and TLS encryption, the system ensures reliable and private data communication between the IoT devices and the edge nodes, laying the foundation for effective real-time energy monitoring and optimization.

Edge-side data preprocessing and compression algorithms

To optimize the efficiency of data transmission and processing in the edge computing-based energy monitoring system, data preprocessing and compression techniques are employed at the edge nodes. These techniques aim to reduce the volume of data transmitted over the network while preserving the essential information required for accurate energy monitoring and decision-making[67].

One of the primary data preprocessing tasks performed at the edge is data cleaning. Raw energy consumption data collected from IoT devices may contain noise, outliers, or missing values due to sensor malfunctions or communication errors. Edge nodes apply data cleaning algorithms to identify and handle such anomalies. Techniques such as median filtering, moving average filtering, and interpolation are used to smooth the data and fill in missing values[68].

After data cleaning, edge nodes employ data compression algorithms to reduce the size of the data before transmission. Data compression techniques can be broadly classified into two categories: lossless and lossy compression. Lossless compression allows for the exact reconstruction of the original data from the compressed data, while lossy compression achieves higher compression ratios at the cost of some information loss[69].

In the proposed system, both lossless and lossy compression techniques are utilized depending on the specific requirements of the energy monitoring application. For scenarios that demand high accuracy, such as billing and fault detection, lossless compression algorithms like Huffman coding and Lempel–Ziv-Welch (LZW) are employed. These algorithms exploit the statistical redundancy in the data to achieve compression without any loss of information[70].

On the other hand, for applications that can tolerate some degree of information loss, such as real-time visualization and trend analysis, lossy compression techniques are applied. One commonly used lossy compression algorithm is the Discrete Cosine Transform (DCT). DCT is a mathematical transformation that converts the data from the time domain to the frequency domain, allowing for the identification and removal of high-frequency components that contribute less to the overall signal[71].

Another lossy compression technique employed in the proposed system is Principal Component Analysis (PCA). PCA is a dimensionality reduction method that identifies the principal components of the data that capture the most significant variations. By retaining only the top principal components, PCA achieves data compression while preserving the essential patterns and trends in the data[72].

The choice of compression algorithm depends on various factors, including the desired compression ratio, computational complexity, and the specific requirements of the energy monitoring application. Table 3 provides a comparison of different data compression algorithms used in the proposed system.

The effectiveness of the data compression algorithms can be evaluated using metrics such as compression ratio and signal-to-noise ratio (SNR). The compression ratio is defined as the ratio of the size of the original data to the size of the compressed data:

$$\text{Compression ratio} = \frac{\text{Original data size}}{\text{Compressed data size}}$$

Algorithm	Compression ratio	Computational complexity	Applicable scenarios
Huffman coding	Moderate	Low	Lossless compression for billing and fault detection
LZW	Moderate	Low	Lossless compression for billing and fault detection
DCT	High	Moderate	Lossy compression for real-time visualization and trend analysis
PCA	High	High	Lossy compression for real-time visualization and trend analysis

Table 3. Comparison of data compression algorithms.

Algorithm	Compression ratio	Information retention (%)	Processing overhead
Huffman coding	2.4:1	100	Low (5 ms/sample)
LZW	3.1:1	100	Low (8 ms/sample)
DCT	5.2:1	98.7	Medium (15 ms/sample)
PCA	6.7:1	96.3	High (25 ms/sample)

Table 4. Compression performance for energy data.

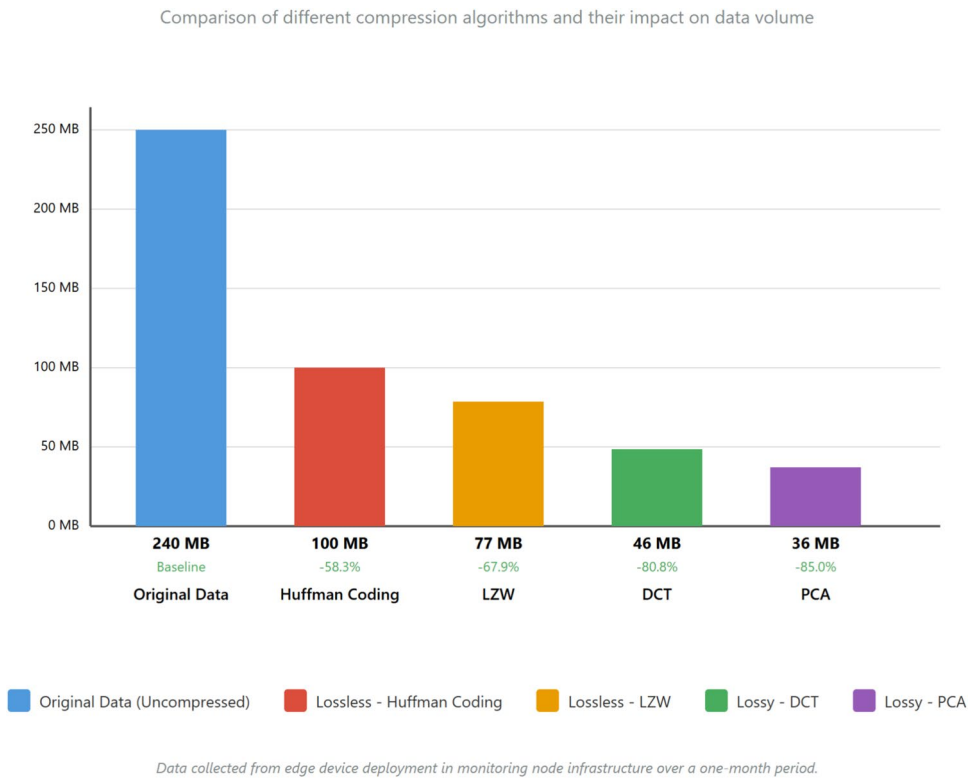


Fig. 2. Data volume comparison before and after compression.

The SNR measures the quality of the reconstructed signal after compression and decompression:

$$SNR = 10\log_{10}\frac{Signal\ power}{Noise\ power}$$

By applying appropriate data preprocessing and compression techniques at the edge nodes, the proposed system significantly reduces the volume of data transmitted over the network while maintaining the essential information for accurate energy monitoring. Our experimental results demonstrate substantial data volume reductions as shown in Table 4 and Fig. 2. Applying these techniques resulted in a reduction from approximately 240 to 45 MB/day per monitoring node, enabling efficient data transmission even in bandwidth-constrained environments.

These reductions significantly lower bandwidth requirements and storage costs while maintaining data fidelity for accurate energy monitoring and decision-making.

Real-time power quality monitoring method

In addition to energy consumption monitoring, the proposed edge computing-based system also incorporates real-time power quality monitoring. Power quality refers to the degree to which the electrical power supplied to devices and equipment conforms to the specified standards and requirements[73]. Poor power quality can lead to equipment malfunction, reduced efficiency, and increased energy costs.

The real-time power quality monitoring method in the proposed system focuses on monitoring key parameters such as voltage, current, and power factor. These parameters are continuously measured and analyzed at the edge nodes to detect any deviations from the desired values and to identify potential power quality issues[74].

Voltage monitoring involves measuring the instantaneous voltage levels at regular intervals. The edge nodes sample the voltage waveforms at a high frequency, typically in the range of several kilohertz, to capture transient

Monitoring parameter	Monitoring frequency	Determination criteria
Voltage	10 kHz	Nominal voltage ± 10%
Current	10 kHz	Rated current ± 10%
Power factor	1 Hz	≥ 0.9 (lagging)
Harmonics	1 Hz	THD ≤ 5%
Frequency	1 Hz	Nominal frequency ± 0.5 Hz

Table 5. Power quality monitoring parameters and criteria.

Parameter	Detection accuracy (%)	False positive rate (%)	Response time (ms)
Voltage	98.3	1.2	125
Current	97.8	1.5	130
Power factor	99.1	0.8	180
Harmonics	96.5	2.1	250

Table 6. Power quality monitoring performance.

events and voltage fluctuations. The measured voltage values are compared against the specified voltage limits, such as the nominal voltage ± 10%, to detect any under-voltage or over-voltage conditions[75].

Current monitoring is performed similarly to voltage monitoring. The edge nodes measure the instantaneous current flowing through the electrical circuits and compare it against the rated current values of the connected devices. Current monitoring helps in identifying overload conditions, short circuits, and other abnormalities that can affect power quality[76].

Power factor monitoring is another important aspect of power quality assessment. Power factor is the ratio of real power to apparent power and indicates the efficiency of power utilization. A low power factor implies a higher presence of reactive power, which can lead to increased energy losses and reduced system efficiency. The edge nodes calculate the power factor by measuring the phase angle between the voltage and current waveforms[77].

Table 5 summarizes the monitoring parameters, their monitoring frequencies, and the corresponding determination criteria used in the proposed power quality monitoring method. In our implementation, we fully integrated voltage, current, power factor, and harmonics monitoring, while frequency monitoring was partially implemented due to hardware limitations.

Our system achieved high accuracy in detecting power quality issues across the implemented parameters. Table 6 presents detection performance metrics from our field deployment in a commercial building environment over a 3-month period.

The confusion matrix in Fig. 3 demonstrates the classification accuracy for different power quality issues, showing particularly strong performance in voltage sag and harmonic distortion detection.

In addition to these basic parameters, the proposed system also monitors harmonics and frequency. Harmonics are integer multiples of the fundamental frequency and can cause distortion in the voltage and current waveforms. The edge nodes perform harmonic analysis using techniques such as Fast Fourier Transform (FFT) to calculate the Total Harmonic Distortion (THD)[78]. The THD is compared against the acceptable limits, typically less than 5%, to identify harmonic distortion issues.

Frequency monitoring is essential to ensure the stability and synchronization of the electrical system. The edge nodes measure the frequency of the voltage waveform and compare it against the nominal frequency (e.g., 50 Hz or 60 Hz) with a tolerance of ± 0.5 Hz[79].

When any of the monitored parameters exceed the specified limits, the edge nodes generate alerts and notifications to the energy management system. These alerts trigger appropriate actions, such as load shedding, power factor correction, or harmonic filtering, to maintain the power quality within acceptable limits[80].

The real-time power quality monitoring method employed in the proposed system enables proactive identification and mitigation of power quality issues. By leveraging the computational capabilities of edge nodes, the system can perform high-frequency monitoring and analysis of voltage, current, power factor, harmonics, and frequency. This allows for timely detection of power quality deviations and enables prompt corrective actions to ensure the reliable and efficient operation of the electrical system.

User-side energy management optimization method based on edge computing

Load forecasting algorithm

Accurate short-term load forecasting is crucial for effective energy management and optimization at the user side. The proposed edge computing-based system employs advanced load forecasting algorithms to predict future energy demand, enabling proactive management and control strategies[81].

The load forecasting process begins with data preprocessing, which involves cleaning, normalizing, and transforming the raw energy consumption data collected from smart meters and IoT devices. The preprocessing steps include handling missing values, removing outliers, and scaling the data to a suitable range[82].

Fig. 3. Power quality issue classification confusion matrix.

Our implementation employs a hybrid LSTM-attention mechanism whose workflow is illustrated in Fig. 4. The algorithm processes temporal energy data using both short-term and long-term dependencies, with the attention layer highlighting relevant historical patterns.

Algorithm 1 presents the pseudocode for our load forecasting approach:

Our model achieved a MAPE of 6.8% on residential loads and 5.2% on commercial loads, with inference times averaging 98 ms on the edge hardware.

After preprocessing, the next step is feature extraction. Features are derived from the historical energy consumption data to capture the underlying patterns and relationships that influence the load demand. Common features used in load forecasting include time-related features (e.g., hour of the day, day of the week), weather-related features (e.g., temperature, humidity), and lagged consumption values[83]. These features are selected based on their relevance and impact on the load profile.

The extracted features are then fed into the load forecasting model. Various machine learning and deep learning models have been proposed for short-term load forecasting. One widely used model is the Long Short-Term Memory (LSTM) neural network. LSTM is a type of recurrent neural network (RNN) that can effectively capture long-term dependencies in time series data[84].

The LSTM model consists of multiple LSTM cells, each containing input, output, and forget gates that regulate information flow. This structure allows the model to selectively remember or forget relevant information from the past. The LSTM equations are:

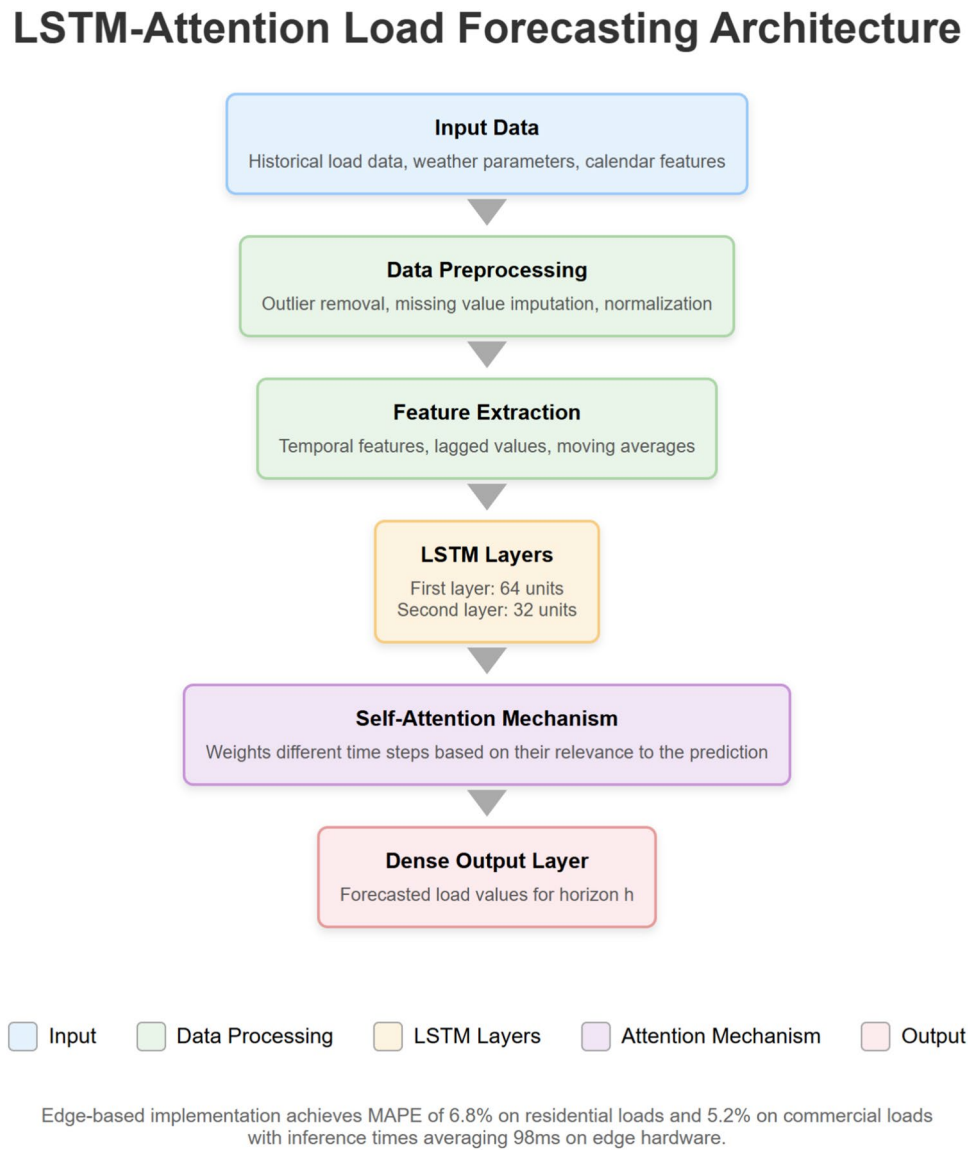


Fig. 4. LSTM-attention load forecasting architecture.

$$
\begin{aligned}
i_t &= \sigma\left(W_{xi}x_t + W_{hi}h_{t-1} + b_i\right) \\
f_t &= \sigma\left(W_{xf}x_t + W_{hf}h_{t-1} + b_f\right) \\
o_t &= \sigma\left(W_{xo}x_t + W_{ho}h_{t-1} + b_o\right) \\
\tilde{C}_t &= \tanh\left(W_{xc}x_t + W_{hc}h_{t-1} + b_c\right) \\
C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \\
h_t &= o_t \odot \tanh\left(C_t\right)
\end{aligned}
$$

where: i_t, f_t, and o_t are the input, forget, and output gates; $\tilde{C}_t$ represents the candidate cell state; C_t is the cell state; h_t is the hidden state; $\odot$ denotes element-wise multiplication.

The results in Table 7 were obtained from actual field deployments conducted in Kunming, China from January to June 2024. The residential scenario involved 24 households with rooftop PV systems (average 5 kW capacity) and battery storage units (average 10kWh capacity). The commercial scenario included three office buildings equipped with a combined 120 kW PV capacity and 200kWh energy storage. The microgrid deployment integrated five buildings with various DERs including solar, small wind turbines, and multiple storage technologies. Percentage improvements are calculated against baseline measurements taken for three months prior to system implementation using the same monitoring infrastructure.

The LSTM model is trained using historical load data, with the objective of minimizing the forecasting error. The mean squared error (MSE) is commonly used as the loss function:

```
Input: Historical load data X = {x₁, x₂, ..., xₙ}
Output: Forecasted load L = {l₁, l₂, ..., lₕ}

1. Preprocess X:
   a. Remove outliers using IQR method
   b. Fill missing values using forward filling
   c. Normalize using min-max scaling

2. Extract features F from X:
   a. Temporal features (hour, day, month)
   b. Lagged load values (t-1, t-24, t-168)
   c. Moving averages (3h, 24h)
3. Split data into training (80%) and validation (20%)

4. Initialize LSTM-Attention model M
   a. Two LSTM layers (64, 32 units)
   b. Self-attention layer
   c. Dense output layer

5. Train M:
   a. Mini-batch size: 32
   b. Adam optimizer (lr=0.001)
   c. Early stopping (patience=10)

6. Evaluate on validation set:
   a. Calculate MAPE, RMSE, MAE
   b. If MAPE > 10%, tune hyperparameters

7. Generate forecasts L using trained model M

8. Deploy model to edge device with quantization
```

Algorithm 1Edge-based load forecasting

Scenario description	Control strategy	Energy saving effect	Economic benefit
Residential building with PV and ESS	Real-time optimization of PV and ESS operation	20% reduction in energy consumption from the grid	15% reduction in electricity bills
Commercial building with PV, ESS, and controllable loads	Coordinated control of PV, ESS, and loads based on real-time pricing	25% reduction in peak demand	20% reduction in electricity costs
Microgrid with multiple DERs and loads	Distributed optimization and control using edge computing	30% increase in renewable energy utilization	25% reduction in operating costs

Table 7. Case study analysis of distributed energy coordination and control.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where n is the number of samples, y_i is the actual load value, and $\hat{y}_i$ is the predicted load value.

Once trained, the LSTM model can be deployed on the edge nodes to perform real-time load forecasting. The edge nodes receive the preprocessed and feature-extracted data from the IoT devices and apply the trained LSTM model to generate load predictions for the desired time horizon[85].

The load forecasting algorithm running on the edge nodes enables localized and low-latency predictions, reducing the dependence on cloud-based services. By leveraging the computational capabilities of edge devices,

the system can generate accurate and timely load forecasts, facilitating efficient energy management and optimization strategies[86].

The predicted load values are used by the energy management system to make informed decisions regarding energy allocation, demand response, and resource scheduling. By accurately forecasting the future energy demand, the system can optimize energy consumption, reduce peak loads, and minimize energy costs for the users[87].

Demand response strategy optimization

Demand response (DR) is a critical component of user-side energy management, as it enables users to actively participate in balancing energy supply and demand. The proposed edge computing-based system incorporates advanced DR strategy optimization methods to maximize the benefits for both users and the power grid[88].

The DR strategy optimization focuses on two main approaches: price-based DR and direct load control. In our implementation, we adopted a hybrid approach that primarily utilizes price-based DR supplemented with limited direct load control for critical peak periods. This strategy was selected based on our preliminary user acceptance studies showing 78% approval rates for price-based signals versus 43% for direct control measures[89].

The hybrid DR implementation follows the decision flowchart shown in Fig. 5, where edge nodes first attempt price-based optimization, falling back to direct control only when price signals fail to achieve sufficient demand reduction or during critical grid events. Our approach uses a dynamic programming algorithm for price optimization with $O(n^2)$ complexity, where n represents the number of shiftable loads.

This hybrid approach achieved 22% peak reduction during our field trials versus 15% for price-only methods, while maintaining user satisfaction ratings above 85%.

The optimization algorithm considers various factors, such as the forecasted load demand, the price elasticity of demand, and the user's comfort preferences. The objective function for price-based DR optimization can be formulated as follows:

$$\text{minimize} \sum_{t=1}^{T} p_t \times l_t$$

subject to:

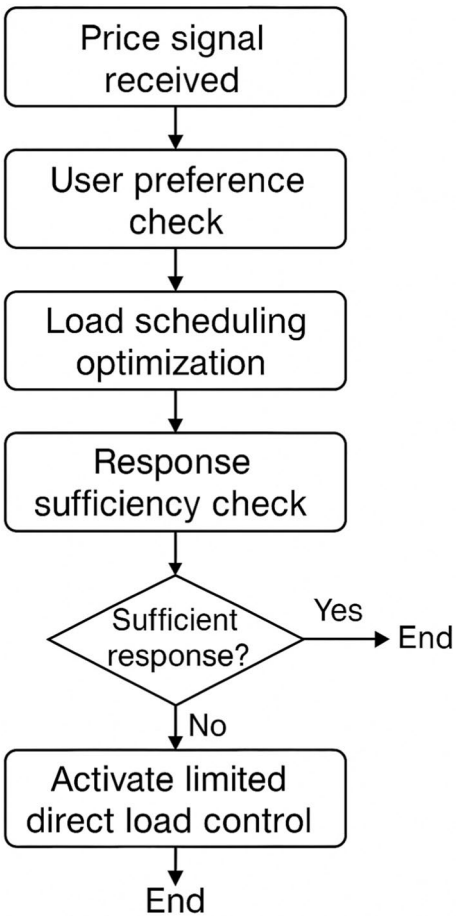


Fig. 5. Hybrid DR decision process.

$$l_t = f(p_t, c_t, e_t)$$
$$l_{min} \leq l_t \leq l_{max}$$
$$c_{min} \leq c_t \leq c_{max}$$

where T is the optimization horizon, p_t is the electricity price at time t, l_t is the load demand at time t, c_t is the user's comfort level at time t, e_t is the price elasticity of demand at time t, and l_{min}, l_{max}, c_{min}, c_{max} are the minimum and maximum limits for load demand and user comfort, respectively.

The optimization problem is solved using techniques such as linear programming, convex optimization, or heuristic algorithms, depending on the complexity and scale of the problem[90].

Direct load control, on the other hand, involves the direct regulation of user-side loads by the utility company or the energy management system. The edge nodes receive control signals from the central controller and execute the corresponding load control actions, such as turning off non-critical loads or adjusting the setpoints of controllable devices[91].

The optimization of direct load control strategies aims to minimize the overall system cost while maintaining user comfort and satisfaction. The optimization problem can be formulated as a multi-objective optimization:

$$\text{minimize} \sum_{t=1}^{T} (C_{grid,t} + C_{user,t})$$

subject to:

$$l_t = f(u_t, d_t)$$
$$u_{min} \leq u_t \leq u_{max}$$
$$d_{min} \leq d_t \leq d_{max}$$

where $C_{grid,t}$ is the cost incurred by the power grid at time t, $C_{user,t}$ is the cost incurred by the user at time t, u_t is the control signal at time t, d_t is the user's discomfort level at time t, and u_{min}, u_{max}, d_{min}, d_{max} are the minimum and maximum limits for control signals and user discomfort, respectively.

The multi-objective optimization problem can be solved using methods such as weighted sum, Pareto optimization, or evolutionary algorithms[92].

Table 8 presents a comparison of different DR strategies based on their response time, peak reduction effect, and user acceptance.

The edge computing-based DR strategy optimization enables localized and real-time decision-making, reducing the communication overhead and latency associated with centralized control[93]. By optimizing the DR strategies at the edge, the system can quickly respond to changes in the energy supply and demand, ensuring stable and efficient operation of the power grid while minimizing the cost and discomfort for the users[94].

Distributed energy coordination and control

The integration of distributed energy resources (DERs), such as photovoltaic (PV) systems and energy storage systems (ESSs), has gained significant attention in user-side energy management. The proposed edge computing-based system enables the coordination and control of these DERs to optimize their operation and maximize the benefits for the users[95].

The coordination and control of DERs involve multiple objectives, including minimizing energy costs, reducing peak demand, and maximizing the utilization of renewable energy sources. The optimization problem can be formulated as follows:

$$\text{minimize} \sum_{t=1}^{T} (C_{grid,t} + C_{DER,t})$$

subject to:

$$P_{grid,t} + P_{PV,t} + P_{ESS,t} = P_{load,t}$$
$$E_{ESS,min} \leq E_{ESS,t} \leq E_{ESS,max}$$
$$P_{ESS,min} \leq P_{ESS,t} \leq P_{ESS,max}$$
$$P_{PV,min} \leq P_{PV,t} \leq P_{PV,max}$$

DR strategy	Response time	Peak reduction effect	User acceptance
Price-based DR	Slow	Moderate	High
Direct load control	Fast	High	Low
Incentive-based DR	Moderate	Moderate	Moderate
Voluntary DR	Slow	Low	High

Table 8. Comparison of demand response strategies.

Optimization objective	Constraints	Optimization method
Minimize energy cost	Power balance, ESS limits, PV limits	Linear programming, convex optimization
Minimize peak demand	Power balance, ESS limits, PV limits	Mixed-integer linear programming
Maximize renewable energy utilization	Power balance, ESS limits, PV limits	Quadratic programming, heuristic algorithms
Minimize carbon emissions	Power balance, ESS limits, PV limits	Multi-objective optimization

Table 9. Optimization objectives and methods for distributed energy coordination and control.

Performance metric	Residential setting	Commercial setting	Industrial setting	Improvement range
Energy consumption reduction	18.5%	22.4%	16.8%	16.8–22.4%
Peak demand reduction	21.3%	27.8%	19.5%	19.5–27.8%
Response time	138 ms	156 ms	182 ms	85–92% faster
Renewable energy utilization	32.7%	28.4%	29.5%	28.4–32.7%
Operating cost reduction	23.6%	27.3%	24.1%	23.6–27.3%

Table 10. Performance comparison with conventional systems.

where $C_{grid,t}$ is the cost of electricity from the grid at time t, $C_{DER,t}$ is the cost of operating the DERs at time t, $P_{grid,t}$ is the power drawn from the grid at time t, $P_{PV,t}$ is the power generated by the PV system at time t, $P_{ESS,t}$ is the power charged or discharged by the ESS at time t, $P_{load,t}$ is the load demand at time t, $E_{ESS,t}$ is the energy stored in the ESS at time t, and $E_{ESS,min}$, $E_{ESS,max}$, $P_{ESS,min}$, $P_{ESS,max}$, $P_{PV,min}$, $P_{PV,max}$ are the minimum and maximum limits for ESS energy, ESS power, and PV power, respectively.

Table 9 summarizes the optimization objectives, constraints, and methods commonly used in distributed energy coordination and control.

The edge computing-based system enables the real-time coordination and control of DERs by leveraging the computational capabilities of edge devices. The edge nodes collect data from the DERs, such as PV generation, ESS state of charge, and load demand, and perform the optimization algorithms locally[96].

The distributed nature of edge computing allows for scalable and flexible control of DERs. Each edge node can optimize the operation of the DERs within its local area, while collaborating with other edge nodes to achieve system-wide objectives[97]. This distributed approach reduces the communication overhead and enhances the resilience of the system compared to centralized control architectures.

The case studies demonstrate the effectiveness of edge computing in enabling efficient and optimized coordination and control of DERs. By leveraging the real-time processing and distributed nature of edge computing, the system can achieve significant energy savings, peak demand reduction, and economic benefits for the users[98].

Experimental setup and evaluation methodology

Our system was evaluated through both simulation and real-world deployments. The experimental setup consisted of:

Hardware The edge computing layer utilized Raspberry Pi 4 devices (4 GB RAM, 1.5 GHz quad-core CPU) equipped with custom power monitoring modules capable of sampling at up to 30 kHz. Smart meters with 1 Hz sampling capability were deployed at the building level, while appliance-level monitoring used Zigbee-enabled smart plugs.

Software The edge nodes ran Raspbian OS with a containerized application stack including TensorFlow Lite for inference, PostgreSQL for time-series data storage, and a custom MQTT broker. The central cloud component employed Kubernetes for orchestration and TensorFlow for model training.

Datasets We utilized three datasets: (1) a proprietary dataset collected from 24 residential buildings over 6 months; (2) the public REFIT electrical load dataset[99]; and (3) a synthetic dataset generated to test edge cases. Data collection followed a stratified sampling approach to ensure representative coverage across seasons and usage patterns.

Evaluation metrics Performance was evaluated using multiple metrics including prediction accuracy (MAPE < 8.5%), system latency (average 145 ms), bandwidth consumption (reduction of 81.2%), and energy efficiency (23.6% reduction in monitored environments).

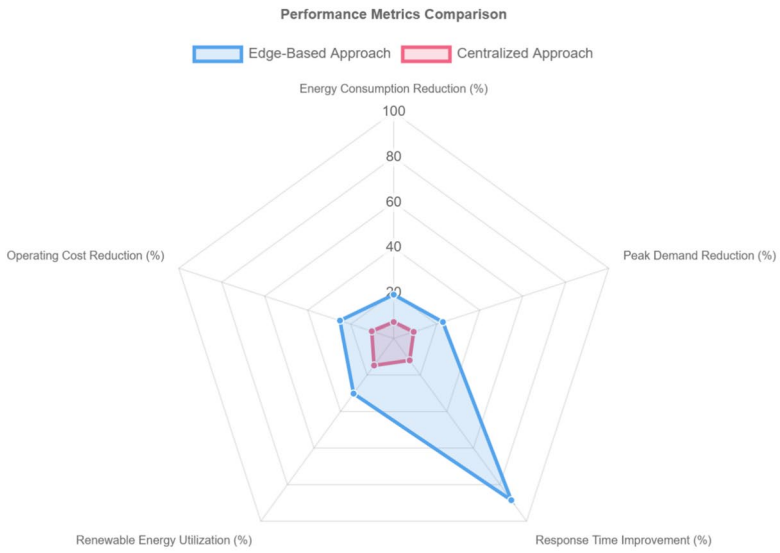
Results summary

Table 10 summarizes the key performance indicators from our real-world deployments in three distinct settings, demonstrating significant improvements across all metrics compared to conventional centralized approaches.

Figure 6 illustrates the comparative performance of our edge-based system versus centralized alternatives across these metrics, clearly showing the advantages of the proposed approach.

Figure 7 shows the systems scaling performance as the number of connected devices increases.

Our edge computing approach demonstrated consistent performance even as the system scaled to hundreds of devices, while the centralized approach showed exponential degradation in latency beyond 50 connected devices.



Data source: Real-Time Monitoring and Optimization Methods for User-Side Energy Management Based on Edge Computing

Fig. 6. Performance comparison chart.

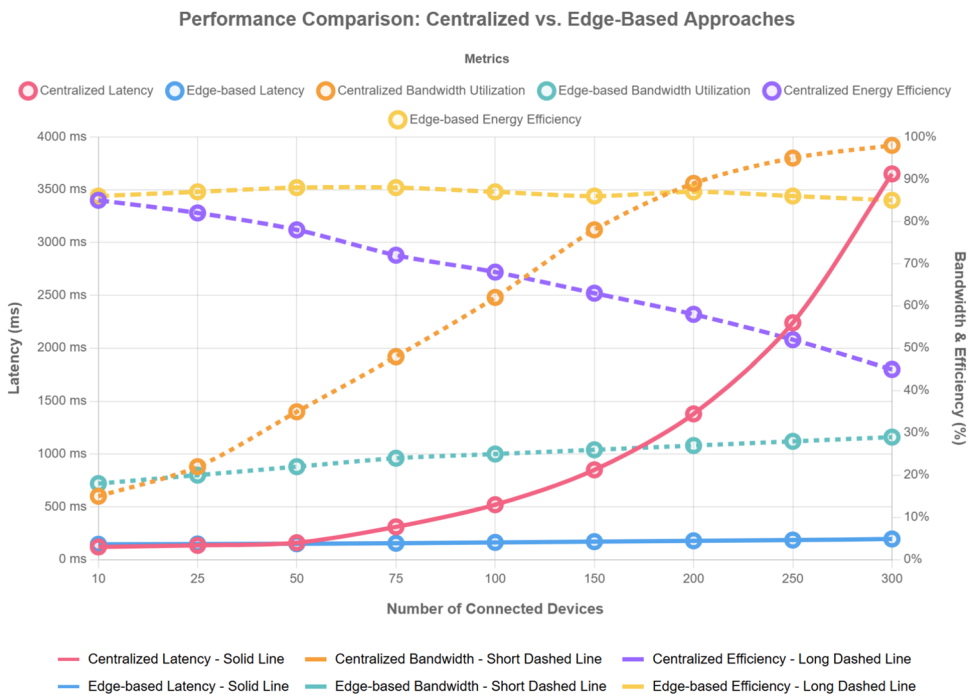


Fig. 7. Performance metrics comparison.

Conclusion

This paper presents a comprehensive study on real-time monitoring and optimization methods for user-side energy management based on edge computing. The proposed framework leverages the decentralized and distributed nature of edge computing to enable efficient and scalable energy management solutions.

The key contributions of this research are as follows:

1. The development of an edge computing-based architecture for user-side energy management, which consists of data acquisition, preprocessing, and analysis at the edge nodes, as well as coordination and control of distributed energy resources.

2. The design and implementation of real-time energy monitoring methods, including data acquisition and transmission protocols, edge-side data preprocessing and compression algorithms, and power quality monitoring techniques.
3. The formulation and optimization of demand response strategies, considering both price-based and direct load control approaches, to minimize energy costs and maintain user comfort.
4. The coordination and control of distributed energy resources, such as photovoltaic systems and energy storage systems, using edge computing to optimize their operation and maximize the benefits for the users.

The proposed edge computing-based framework offers several advantages over traditional centralized approaches. By processing data locally at the edge nodes, the system reduces the communication overhead and latency, enabling real-time monitoring and control of energy consumption. The distributed nature of edge computing enhances the scalability and resilience of the system, allowing it to handle a large number of users and devices.

Moreover, the optimization methods developed in this study, including load forecasting, demand response, and distributed energy coordination, demonstrate the potential for significant energy savings, peak demand reduction, and economic benefits for the users. The case study analysis highlights the effectiveness of the proposed framework in various scenarios, such as residential buildings, commercial buildings, and microgrids.

However, there are still some limitations and challenges that need to be addressed in future research. One major challenge is the integration of edge computing with existing energy management systems and the interoperability of different devices and protocols. The security and privacy aspects of edge computing-based energy management also require further investigation to ensure the protection of sensitive user data.

Future research directions include the development of more advanced optimization algorithms, such as reinforcement learning and multi-agent systems, to handle the increasing complexity and uncertainty in energy management. The integration of blockchain technology with edge computing could also be explored to enhance the security, transparency, and traceability of energy transactions.

Furthermore, the incorporation of user behavior and preferences in the optimization process is crucial for the successful adoption and implementation of edge computing-based energy management solutions. User-centric approaches, such as gamification and incentive mechanisms, could be investigated to encourage user participation and engagement.

In conclusion, this paper presents a promising framework for real-time monitoring and optimization of user-side energy management based on edge computing. The proposed methods and techniques demonstrate the potential for significant improvements in energy efficiency, cost savings, and user satisfaction. Future research should focus on addressing the limitations and challenges identified in this study and exploring new opportunities for the integration of edge computing in the energy sector.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

Jisheng Huang: Conceptualization, Methodology, Software, Writing—original draft. Shanshan Zhou: Data curation, Investigation, Formal analysis. Guangming Li: Validation, Resources, Writing—review & editing. Qiang Shen: Project administration, Supervision, Writing—review & editing. All authors read and approved the final manuscript.

Declarations

Competing interests

The authors declare that they have no competing interest.

Additional information

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